Rahul Pareek

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TECHNICAL SKILLS

Programming Languages	: Python, JavaScript
Frontend	: NextJS, ReactJS
Web Technologies / Frameworks	: FastApi, Django, Flask, Node.js, Express.js, Prisma
AI/LLM	: Google ADK, AWS Strands, VertexAI, CCAI, LangGraph, LangChain
Databases & Infra	: MongoDB, PostgreSQL, Elasticsearch, Vector DBs, AWS, GCP, Azure
Tools & Libraries	: Docker, Kubernetes, Redis

EXPERIENCE

Sr. Member Technical — Nov 2025 – Current

ADP

- Engineered a **distributed Python PDF generation pipeline**, replacing legacy batch workflows with parallel worker queues and horizontal scaling, improving document generation and signing throughput by **40%**.
- Contributed to a **Simplified MCP architecture** using a shared endpoint, **Auth** → **Discover** → **Execute** three tool model and progressive schema disclosure reducing upfront agent context by **up to 98.5%** (~60K to ~900 tokens) while enabling reuse of existing REST APIs.
- **Led development of ADP Assist**, an enterprise AI assistant serving **2K+ users** leveraging **AWS Strands Agents, Amazon Bedrock, AgentCore and MCP** for agent orchestration, conversational state management, enterprise API integration, and automated escalation.
- Designed and implemented an **enterprise RAG pipeline** integrating **Amazon Neptune knowledge graphs and OpenSearch vector retrieval**, enabling graph guided document discovery, semantic search, and context-aware responses across enterprise knowledge sources.
- Owned the architecture and development of an **MCP Certification Engine**, automating API certification and test-collection generation to standardize validation and accelerate MCP onboarding.
- Developed **AI Studio**, an enterprise AI agent platform supporting agent creation, registration, certification, and deployment through an **AWS Hub Spoke architecture**, with centralized governance, tenant isolation, and multi-environment promotion.
- Implemented **automated AI agent evaluation** using **Jenkins, Arize and golden datasets**, enabling repeatable agent testing, response-quality assessment, and performance monitoring.
- Led modernization of legacy **Java microservices**, migrating workloads to **AWS EKS** and replacing RabbitMQ-based processing with **Amazon SQS asynchronous workflows**, improving scalability and operational efficiency.

Sr. Software Engineer — May 2025 – Oct 2025

Google *(Contract)*

- Designed and implemented **Conversational AI and NLP** capabilities for inquiry platform chatbot including Gen AI fallback mechanisms on **Google CCAI(Dialogflow)**. Leveraged **data stores to use user-manuals for RAG based system**, improving response quality for unmatched intents and reducing fallback failure rates.
- Developing versatile API enabling **CRUD operations** on Excel sheets for sales target data retrieval from in-house APIs enabling fast delivery and data updating using **Google Cloud Run**.
- **Resolved critical webhook data point issues** that prevented data processing ensuring seamless communication between the chatbot and backend systems. Integrated bot onto website UI.
- Optimized pricing recompilation component performance by **setting rule based pricing and using TTL**.

Freelance Contractor — Feb 2025 - April 2025

- Developed a Standalone Streamlit app for running energy simulation platform integrating DXF geometry processing with EnergyPlus, automating building-model generation, zoning, floor/plenum configuration, WWR definition, and simulation execution with result management.
- Engineered an automated multi-model simulation pipeline to execute multiple EnergyPlus IDF models against weather data, validate simulation health through error-log analysis, and surface success/failure diagnostics.
- Implemented simulation lifecycle and results management, including project-based storage, simulation history, output file discovery, individual file downloads, and consolidated ZIP exports.

Software Engineer — Feb 2022 – Feb 2025

EdgeVerve

- Developed Scalable RESTful API for Finacle, a global financial platform utilized by over **150 banks in more than 100 countries**, that serves a user base of over **1.7 billion account holders.**
- **Migrated the database from MySQL to PostgreSQL. Conducted load testing on APIs** to ensure efficient job processing under high load after integrating Kafka queuing.
- Enhanced internal data hydration tool by adding support for JSON and CSV data types to improve usability in current code**.**
- Optimized JSON file parsing in the Finacle core component by adopting a more efficient approach utilizing parallel processing of data along with Json Parser to parse the file. This enhancement led to a **34% reduction in the overall runtime of the component during daily performance runs**.
- Debugged and fixed issues with Kubernetes (GKE) that lead to Pods not getting data under load added additional quality checks in Jenkins pipeline and **improved the test coverage from 45% to 80%** for Finacle core component.
- Implemented c**entralized JSON file access for core components,** thereby reducing method calls along with removing redundant code. These efforts led to **20% improvement in performance runs**.
- Spearheaded WhatsApp Banking Chatbot Project Organized and re-grouped existing flow on basis of categories for ease of access **fixed performance bottlenecks while** parsing pdf under high load, **added rate limiting** to prevent API abuse. Migrate FMT Ui from low code to react worked on optimizing record reads.
- Integrated DialogFlow with WhatsApp banking chatbot for LLM Integration with different flow for acc options and support along with TTS support using smallest.ai.

EDUCATION

Rajasthan Technical University
B.Tech (Computer Science)